COLLEGE OF ENGINEERING PUNE

(An Autonomous Institute of Government of Maharashtra.)

END Semester Examination

Programme: B.Tech

Semester: V

Course Code: 21001

Course Name: Database Management System

Branch: Computer Engineering

Academic Year: 2022-23

Duration: 3 hours

Max Marks: 60

1	1	2	0	0	3	1	1	4

Student PRN No. Instructions:

1. **Figures to the right indicate the full marks.**
2. **Mobile phones and programmable calculators are strictly prohibited.**
3. **Writing anything on question paper is not allowed.**
4. **Exchange/Sharing of stationery, calculator etc. not allowed.**
5. **Write your PRN Number on Question Paper.**

<table>
<tr><th></th><th></th><th></th><th>Marks</th><th>CO</th><th>PO</th></tr>
<tr><td>Q 1</td><td>a</td><td>A university registrar's office maintains data about the following entities: (a) courses, including number, title, credits, syllabus, and prerequisites; (b) course offerings, including course number, year, semester, section number, instructor(s),timings, and classroom; (c) students, including student-id, name, and program; and (d) instructors, including identification number, name, department, and title.

Further, the enrollment of students in courses and grades awarded to students in each course they are enrolled for must be appropriately modeled.

Construct an E-R diagram for the registrar's office. Consider the given mapping constraints: "The entity set course-offering is a weak entity set dependent on course".

Make following assumption: "A class meets only at one particular place and time. This E-R diagram cannot model a class meeting at different places at different times".</td><td>6</td><td>2</td><td>b</td></tr>
<tr><td>Q2</td><td>a</td><td>The following table has two attributes A and C where A is the primary key and C is the foreign key referencing A with on delete cascade. What is the effect of deleting row (2 ,4) from table? Draw table for only data values left in table.
<table>
<tr><th>A</th><th>C</th></tr>
<tr><td>2</td><td>4</td></tr>
<tr><td>3</td><td>4</td></tr>
<tr><td>4</td><td>3</td></tr>
<tr><td>5</td><td>2</td></tr>
<tr><td>7</td><td>2</td></tr>
<tr><td>9</td><td>5</td></tr>
<tr><td>6</td><td>4</td></tr>
</table></td><td>3</td><td>3</td><td>b, g</td></tr>
<tr><td></td><td>b</td><td>Explain with example what is Super key, Candidate key.</td><td>2</td><td></td><td></td></tr>
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Q 3 **a** Consider three tables R(a,b,c), S(d,e,f), and T(g,h,i). For the relational algebra expression, Construct the correct Equivalent SQL query . **3** **3** **b, g**

Note: DO not use "in", "with", "Natural Join" , (cartesian product allowed), "view", "Nested query" and rename operator. Cartesian product is allowed to use.

$\pi_{R.b}\,(\sigma_{Total>2}\,({}_{R.B}\mathcal{G}_{count(*) \rightarrow Total}\,(R \bowtie_{R.a = s.d} ((\sigma_{s.f\ <50}\,(S)) \bowtie_{S.e = T.g} (\sigma_{T.h > 21}\,(T))))))$

b Consider the employee database where the primary keys are underlined. Give an expression in SQL for the following queries. **3**

employee (**<u>employee-name</u>**, street, city)
works (**<u>employee-name</u>**, company-name, salary)
company (**<u>company-name</u>**, city)
manages (**<u>employee-name</u>**, manager-name)

Give all employees of First Bank Corporation a 10-percent raise
Note : Do not use "join" , "as" and " with", "view" clause .

c Consider the employee database as given in **Q3 b** , where the primary keys are underlined. Give an expression in SQL for the following queries. **3**

Find average salary of each company . Output should show company name and its average salary.
Do not use "join" , "as" and " with", "view" clause and nested query.

Q4 **A** Suppose that we ~~decompose~~ have the schema $R = (A, B, C, D, E)$ and following set F of functional dependencies holds: List four candidate keys for R. Show the proof for the same. **4** **4** **b, d**

$A \rightarrow BC$
$CD \rightarrow E$
$B \rightarrow D$
$E \rightarrow A$

B. Analyze if the following decomposition of the schema R is a lossless-join decomposition or lossy decomposition. **4**

R1= (A, B, C)
R2 = (C, D, E). use the following example of r:

A	B	C	D	E
a1	b1	c1	d1	e1
a2	b2	c1	d2	e2

With R1= (A, B, C) R2 = (C, D, E)

C Consider the below given relational schema and set of functional dependencies. **4**

C_B_B = (<u>C, E</u>, B, T) candidate key = (C, E)

F = Fc = {C,E -> B,T, ; E -> B} . Analyze if the given schema is in 3NF ? Is the given schema in BCNF? Provide the reason(s) for the same.
Provide the dependency preserving , lossless 3NF deomposition for the same .

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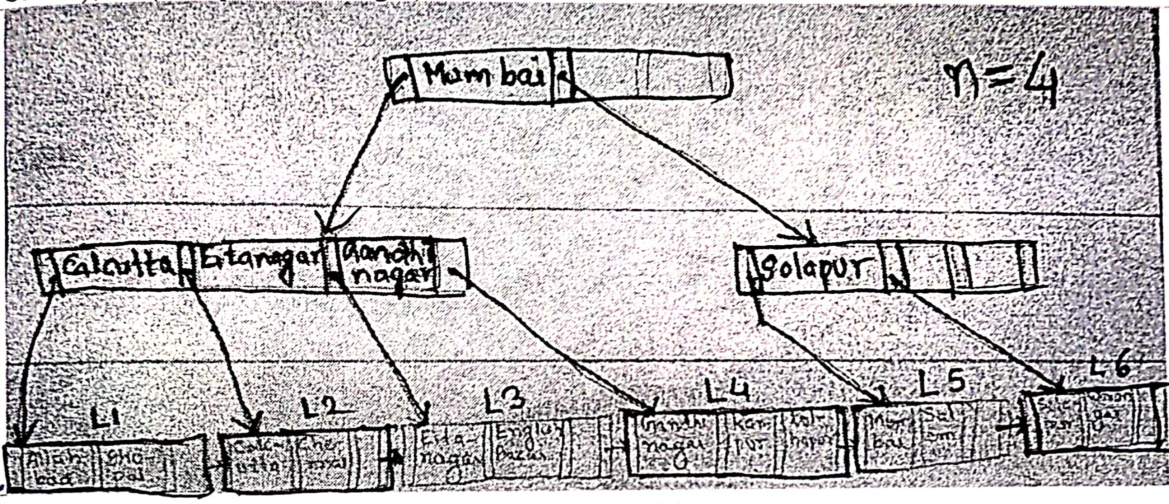
Q5	a	Analyze the two different types of ordered indices with example. Can the Non-Clustered indices be sparse indices ? Why ?	6	5	d
	b.	Consider the given B+ tree index file with key values as city names as shown below. The value of n = 4 in this case. Design the B+ tree obtained after inserting the value of "Latur" in the tree. In case the names are not clear in picture (in leaf node), the names of city are in ascending order: L1(Allahbad, Bhopal), L2(Calcutta, Chennai), L3(Eitanager, English bazaar), L4(Ghandhinagar, Kanpur, Kolhapur), L5(Mumbai, Salem), L6(Solapur, Warangal)  	4		
Q6	a	Consider the following two transactions: T1 : read(A); read(B); if A = 0 then B := B + 1; write(B). T2 : read(B); read(A); if B = 0 then A := A + 1; write(A). Add lock and unlock instructions to transactions T1 and T2 , so that they observe the two-phase locking protocol. Can the execution of these transactions result in a deadlock?	6	6	d , g
	b	Consider two transactions T1 and T2 and four schedules S1 , S2 , S3 , S4 of T1 and T2 as given below. Which of the below schedules are conflict-serializable? Also give its serial equvilaent . T1: R1[x] ; W1[x]; W1[y]; T2: R2[x] ; R2[y]; W2[y]; S1: R1 [x] ; R2 [x]; R2[y]; W1[x]; W1[y]; W2[y] S2: R1 [x] ; R2[x]; R2[y] ; W1[x]; W2[y]; W1[y] S3: R1 [x] ; W1[x] ; R2[x] ; W1[y] ; R2[y] ; W2[y] S4: R2 [x] ; R2[y] ; R1 [x] ; W1[x] ; W1[y]; W2[y]	4		
	c.	With refernce to logbased recovery. State the sequence of action to take place when the system performs <checkpoint>.	4		
	d	Write a short note about recoverable schedule . Provide example for the same	4		

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